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Relating growth and nutrition to site factors in young chestnut plantations established on agricultural and forest land in northern Spain

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Abstract The growth and nutritional status of 97 hybrid chestnut plantations on former agricultural land and forest land were studied in relation to site conditions (climatic variables and soil properties). A single family of curves was obtained for classifying the early height growth of hybrid and sweet chestnut. Soil limitations and nutrient deficiencies were evident in the former forest land, whereas the agricultural soils were found to be generally suitable for growing chestnut. The stands with highest growth rates were characterized by high foliar concentrations of K, P and Ca. Successful growth of the plantations was always related to high summer precipitation, low elevation and absence of long periods of risk of frost. The results confirm the adaptability of chestnut to most sites where it has been planted for timber production.

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Introduction

Chestnut (*Castanea sativa* Mill.) is considered a native species in the Iberian Peninsula (Krebs et al. 2004; Conedera et al. 2004), where systematic cultivation for nut and wood production began as early as the Middle Ages (Conedera et al. 2004) and became a means of subsistence for rural mountain populations. The types of chestnut stand existing at present differ widely in terms of silvicultural treatments and main production aim, and include plantations for nut (and sometimes wood) production, coppice stands, often of ancient origin (Cabrera Bonet 1997), and high forest, mainly planted for timber (Bourgeois 1992).

In the region of Galicia, in northwestern Spain, chestnut trees cover an area of 70,000 ha. The extent of the cover has decreased considerably in past decades as a result of the abandonment of traditional cultural practices, the occurrence of ink disease (*Phytophtora cinnamomi* Rand) (which has also caused extensive damage to chestnut trees in Atlantic areas, especially in coastal sites at low elevations), and the effects of canker (Fernández López 1984; Vieitez Cortizo et al. 1996; Mansilla Vázquez et al. 2000; Álvarez-Álvarez and Rodríguez-Soalleiro 2003).

In response to land abandonment and demographic decline in rural areas, as well as environmental and economic strategies, afforestation of arable land is increasingly carried out in the Iberian Peninsula. Chestnut is readily accepted for planting by private landowners, and the current plantation rate for Portugal, northern Spain and France has been estimated at 4,500 ha year^{-1} (Álvarez-Álvarez 2004). The attraction of chestnut is the prospect of production of valuable hardwood timber in relatively short rotations, as chestnuts have a shorter rotation than other hardwoods (Kerr and Evans 1993).

The use of ink-resistant hybrid chestnut, obtained by crossing *Castanea sativa* Mill., *Castanea crenata* Siebold & Zucc and *Castanea mollissima* Blume was initiated in 1943 by Urquijo (Fernández-López et al. 1993), and several clones have been selected (from 1989 onwards) for their desirable characteristics in terms of timber production (i.e., straightness, branchiness and rapid height growth) at the Lourizán Centre of Forestry and Environment Research (Spain) (Fernández-López et al. 1993, 1995, 2008; Miranda-Fontaiña et al. 2007; Furones-Pérez and Fernández-López 2008). Clonal material with defined characteristics of resistance to *Phytophthora* sp., vigour, stem form and flushing has been approved recently (BOE 2007). The sensitive sweet chestnut is also planted in elevated areas free from the disease. The occurrence of diseases and pests such as *Phytophtora cinnamomi* Rand and *Endothia parasitica* (Murril) Anderson severely hampers the choice of species, selection of planting density and tending operations such as shaping and pruning.

Although fruit production has traditionally been the main goal of management in northwestern Spain, timber production is now the primary production aim, especially on steep and stony ground, where harvesting and selling the fruit is not cost-effective (Álvarez et al. 2000). Several recent studies have shown that plantations destined for fruit production are generally located in areas with low rainfall in summer, moderate water deficit and low slope (Rubio et al. 2002; Gandullo Gutiérrez et al. 2004), although the optimal conditions for timber production usually differ (Bourgeois 1992), and there is still a lack of knowledge about the relationships among early growth, site parameters and tending operations.

In Galicia, most afforestation of agricultural land has been carried out by means of subsidies provided by the local government, within the framework of the Common Agricultural Policy (European Union). Most of the land converted to forest has been grassland and cultivated areas either already abandoned or in the process of being abandoned (Álvarez-Álvarez 2004). As regards plantations on former agricultural land it is not known whether the owners intended to diversify the productive system. In forest land, most plantations have been established by private owners, and only a small proportion of the stands have been planted by forest authorities. The objective in both cases was to transform shrubland into productive forest.

The objective of the present study was to investigate the effect of site variables and nutrient status on height and diameter growth in newly established plantations of chestnut destined for timber production.

Materials and methods

Study area and sampling

The area of study is located in Galicia (northwestern Spain), a region considered as highly suitable for timber growth because of the Atlantic climatic conditions, i.e., mild temperatures, high rainfall and low moisture deficit. The plantations have been established on land formerly under two different types of use: non-wooded land covered mainly by shrubs (hereafter referred to as forest land), and agricultural land. The soils are developed over siliceous substrates, ranging from igneous rocks such as granite, to metamorphic rocks such as shale and slates. The forest soils, which have not been cultivated or intensively fertilized, are moderately deep, have a high proportion of coarse fragments, strong acidity, high organic matter and low nutrient content, mainly due to the strong weathering they have undergone. Most are classified as Leptosols or Regosols (Queijeiro et al. 2000). The texture is loamy and, in the case of soils developed on granite, sandy. The agricultural soils are usually deeper, have less organic matter and greater nutrient content than the forest soils, and most are classified as Dystric Cambisols, Humic Umbrisols or Regosols.

The operations involved in chestnut establishment include clearing, mechanical site preparation and plantation of sweet chestnut (*Castanea sativa* Mill.) seedlings or layers of hybrid chestnut (*Castanea* × *coudercii* A. Camus). The latter is preferred for

plantations at low altitudes, and at least twelve clones are currently used for commercial planting, although they are usually mixed and trees are not identified as belonging to a particular clone. For all plantations sampled, the owners were not aware of the origin of the hybrid chestnut clones used. When plantations were established by owners, the trees were sometimes obtained from third parties, although in other cases the planting was carried out by forest companies. Consequently, it was not possible to determine the origin of the genetic material or the nursery of origin.

The most widely used site preparation technique is subsoiling, and initial stocking usually ranges from 500 to 1,300 plants ha^{-1}, depending on further shaping and respacing operations.

Information about the area planted with chestnut in the last 15 years in Galicia was collected and mapped before selecting the plantations to be studied. With this aim, data from chestnut plantations established within the framework of the Common Agricultural Policy (European Union), and data from other plantations established by the Regional Forestry Administration through forestry enterprises, were used.

The areas were stratified according to the main climatic and soil classifications presently available for Spain (Elena 1996) by use of ArcMap^{TM} 9.2 GIS software (ESRI 2006), and the plots were distributed between the different strata, in order to represent both former agricultural and forest land, as well as a wide range of climatic conditions and soil properties (some of which depend on the previous land use).

A total of 97 young chestnut plantations (*Castanea sativa* and *Castanea* × *coudercii*) were selected for study. The general distribution of the inventoried stands in relation to silvicultural practices and understory vegetation is shown in Table 1. The plantations were 1–16 year old stands managed by forest owners or by the regional Forest Authority. All were destined for timber production, and thus there were no grafted trees and no treatments for fruit production had been carried out. Plantations that had already been respaced or that displayed symptoms of significant damage due to disease were not included.

Usually one plot per plantation was established, although two or more were delimited when the plantation was large and varied in terms of site factors. A total of 106 plots were analysed, of which 67 were established specifically for this study, and another 39 were available from previous studies in which the same methods, except leaf and soil sampling, had been applied. Plots were sited at least 10 m inside the plantation to avoid edge effects and varied in size between 20 × 20 m and 40 × 40 m, with a minimum of 60 trees included in each plot.

Data collection

The following information was recorded in each plot: general data (age, slope and elevation, total area planted), silvicultural data (species, initial spacing and current stocking and site preparation technique) and general soil data (depth, stoniness, parent material). Total height and breast-height diameter were recorded once for all trees.

The climatic data were obtained from maps of climatic variables for the region, in which an observation period of at least 30 years period was applied (1961–1990) (De Uña 2001). A total of 15 climatic variables were determined (Table 2). The temperature amplitude (TA) was calculated as the difference between the mean temperature of the hottest and coldest months (usually January and July). The risk of frost was considered to exist when the minimum temperature fell below 2°C (Oosterbaan 1998). Precipitation deficit was calculated as the sum of monthly differences between evapotranspiration and rainfall. Potential bioclimatic intensity was calculated as the area derived by plotting mean monthly temperature against a baseline of 7.5°C. The range of values obtained (Table 2) were similar to,

Table 1 Distribution of plots in relation to former land use and type of reproductive material used

	Type of plant		Main understory vegetation				
	Seedling	Clonal	Brooms	Bracken-brambles	Gorse	Heather	Grasses
Forest land	23	43	3	10	5	4	2
Agriculture land	8	32	4	8	4	2	25
Total	31	75	7	18	9	6	27

Table 2 Physiographic and climatic conditions

Variables	Code	Mean	Min	Max	St. Dev.
Elevation	ELEVATION	510.9	250	1086	160.7
Slope	SLO	17.2	0	80	18.0
Precipitation deficit (mm)	PD	189.2	75.0	325.0	67.9
Annual total precipitation (mm)	ATP	1350.7	900	1900	263.6
Winter precipitation (mm)	WP	602.2	350	750	132.9
Summer precipitation (mm)	SP	127.6	90	150	23.5
Evapotranspiration (mm)	ETP	642.2	575	725	39.4
Annual number of days with precipitation	ANDWP	135.7	125	160	12.3
Annual number of days without risk of frost	ANDWOF	79.5	25	125	39.6
Annual number of days with risk of frost (0–2°C)	ANDWF	236.2	125	325	48.3
Annual mean temperature (°C)	AMT	11.7	9	14	1.4
Maximum mean temperature (°C)	MAMT	16.7	13.5	20.5	1.7
Minimum mean temperature (°C)	MMT	4.8	2.5	7.5	0.9
Temperature amplitude (°C)	TA	13.70	12.0	16.0	1.35
January mean temperature (°C)	JMT	6.3	2	9	1.5
August mean temperature (°C)	AUMT	18.4	16	22	1.6
Potential bioclimatic intensity (°C)	PBI	10.6	5	14	2.6

but more variable than, those reported for the natural habitat of sweet chestnut in the region (Blanco et al. 2000).

Leaf and soil sampling was carried out in 67 plots selected after the stratification, from a representative subplot measuring 20 × 20 m. For soil sampling, three pits of 40 cm depth were dug in each subplot, and the soil was combined to form a single bulked sample per plot. Soil samples were air-dried and sieved with a 2 mm screen before analysis. The pH was measured in 0.1 M KCl (soil:solution ratio 1:2.5) with a glass electrode. Total C, N and S were analyzed with a LECO Elemental Analyzer. Organic matter was calculated as the amount of C measured, multiplied by 1.72. Available cations (P, K, Ca, Mg, Mn, Fe, B, Zn, Cu and Ni) were extracted by the Mehlich 3 procedure (Mehlich 1984). Phosphorus was determined by the ascorbic acid method, whereas the other elements were analyzed by atomic absorption spectrophotometry.

Unshaded and undamaged leaves were sampled in June from the upper third of the crowns of 30 trees in the subplot. The collected material was divided into two groups of samples and treated separately in the laboratory. The samples were oven-dried (65°C) to a constant weight, milled (0.25 mm) and digested with H_2SO_4/H_2O_2 (Jones et al. 1991). In leaf digestions, K, Ca, Mg, Mn, Fe, Cu, Ni and Zn were analyzed by atomic absorption spectrophotometry, whereas P was determined photometrically. Nitrogen and S in leaves were analyzed in solid milled material, with a LECO analyzer. Evaluation of the nutritional status of the trees was made on the basis of the levels of nutrients in leaves and soils.

Evaluation of tree growth

To evaluate the rate of growth of the young stands, the top height was first calculated as the mean height of the 100 tallest trees ha^{-1} (van Laar and Akça 1997). Mean annual increment in top height was strongly dependent on the age of each plot, and therefore a guide curve was fitted by use of the generally applied Richard's model (Eq. 1):

$$Ho = a \cdot \left(1 - e^{-bt}\right)^{c} \quad (1)$$

where H_0 is the top height, t is the age of the plantation and a, b and c are parameters.

The Richards equation is one of the most common growth functions used in forest research and fulfils a number of desirable attributes (von Gadow and Hui 1999). No other equations were fitted, since the data used was obtained from a single inventory and does

not allow specific study of growth patterns. The change in the asymptotic parameter *a* enables development of fans of anamorphic curves, useful for classifying the early height growth in plantations. A reference age as young as 10 years was chosen for calculating the top height, and this value (H10) was used to determine growth classes in order to relate these to site factors.

To check if the growth patterns differed in the two species, the dummy variable technique was applied, by fitting Eq. 2:

$$Ho = (a + d \cdot I) \cdot \left(1 - e^{-(b+eI)t}\right)^{c+fI} \quad (2)$$

where $I = 0$ for *Castanea sativa* and $I = 1$ for *Castanea* × *coudercii*.

The mean annual increment in diameter (AID) was calculated by dividing the quadratic mean breast height diameter of each plot by the plantation age. Descriptive statistics were used to group the growth variables (H10 and AID) into three different categories: 'optimal' (H), 'suitable' (M) or 'unsuitable' areas (L). The limits established were: L (<6 m), M (6–8 m), H (>8 m) for H10; and L (<0.6 cm year^{-1}), M (0.6–0.8 cm year^{-1}); H (>0.8 cm year^{-1}) for AID.

The criteria used to evaluate the model were the coefficient of determination R^2, adjusted R^2 and the Root Mean Squared Error (RMSE), evaluated from the Prediction Error Sum of Squares (PRESS residuals). The estimated value of each parameter had to be significantly different from zero.

Statistical analysis

Discriminant analysis (DA) was used to determine the linear combinations of quantitative variables that best reveal differences among the growth classes or competing vegetation groups. The discriminant functions were calculated as linear combinations of the explicative variables by use of the following equation in the SPSS statistical package (SPSS 2007):

$$D_i = b_{i0} + b_{i1}X_1 + b_{i2}X_2 + \cdots + b_{ip}X_p \quad (3)$$

where Di is the discriminant score computed for the i group, $X_1,\ldots,X_p$ are the predictor variables, and $b_{i0},\ldots,b_{ip}$ are parameters that are estimated in order to maximize the ratio of between group and within group variances.

Canonical correlation and the chi-square test were used to identify the significant discriminant functions. The standardized discriminant function coefficients were calculated to indicate the partial contribution of each variable to the discriminant function(s).

Results

Growth rates of chestnuts plantations

The results of the parameter estimation, the standard errors and quantitative statistics for the curves of top height growth (Eq. 1) are shown in Table 3. Anamorphic curves, in which the asymptotic parameter (a) is changed, were developed to produce a family of top height curves of 4.5, 6.5, 8.5 and 10.5 m at a reference age of 10 years, graphically showing a good fit to the data (Fig. 1). The results of the dummy variables equation fit (Eq. 2) showed that the same curves can be used to classify the height growth of hybrid and sweet chestnut, since 0 is within the confidence intervals for parameters d, e and f (Table 3).

The values of H10 varied widely, from 1.6 to 11.9 m (mean, 6.9 m), scarcely deviated from the normal distribution, and showed no relationship with age. This reflects the good growth conditions for some plantations, with mean annual growth rates greater

Table 3 Parameter estimates and quantitative statistics for Eqs. 1 and 2

Parameter	Estimate	Std. Error	Confidence interval 95%	
			Lower	Upper
Equation 1				
a	9.663	3.135	3.443	15.883
b	−0.476	3.727	−7.871	6.918
c	0.151	0.154	−0.154	0.458
d	−0.040	0.170	−0.379	0.298
e	0.992	0.522	−0.043	2.029
f	−0.196	0.544	−1.292	0.903

Parameter	Estimate	Std. Error	RMSE	Adj. R^2
Equation 2				
a	8.916	1.278	1.2262	0.7028
b	0.139	0.067		
c	0.886	0.021		

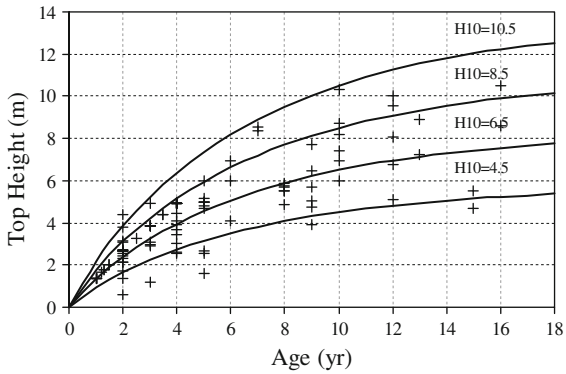


Fig. 1 Top height curves for early age growth of *Castanea* sp. in Galicia

than 1 m. The mean value for agricultural land exceeded the forest land value by 0.6 m, a difference that was only significant at $P = 0.09$. For AID, the total range was also very wide, including plantations that had not reached breast height, and others with an AID as high as 1.57 cm year^{-1}, also with a higher, but not significant, mean for agricultural land.

Nutritional status depending on former land use

The main properties of the soils studied are shown in Table 4. In most of the stands, soils were deeper than 50 cm; shallow soils (<30 cm deep) were found in only three stands. The proportion of coarse elements was higher than 40% in 30% of the soils. Available water content capacity was higher than 100 mm in all cases. Comparison of the previous different types of land use showed that former agricultural soils were characterized by lower organic matter contents, total N and S, and higher P/N ratios (Table 4). Significantly higher concentrations of extractable P, K, Ca, Mg, Fe, Zn and Cu were also observed in these soils, with the differences being attributable to intensive ploughing and fertilization.

In the stands established on previous forest land, the average foliar concentrations of P and N, and the P/N ratios were lower than those established on former agricultural land (Table 5). Concentrations of some micronutrients, such as Zn, Ni and Cu, were also significantly lower. Considering threshold concentrations for deciduous trees of the Fagaceae family (Ende and Evers 1997; Leonardi et al. 1996; González Rodríguez 1975), nutrient deficiencies were more frequent in forest land, in which three cases of low concentrations of P (lower than 1 mg g^{-1}), 9 of Ca (lower than 3 mg g^{-1}), 3 of Mg (lower than 1.6 mg g^{-1}) and 3 of N (lower than 17 mg g^{-1}) were recorded.

The relationship between soil properties and foliar nutrient concentrations were weak. Only foliar Ca

Table 4 Chemical characteristics of the mineral surface horizons of the soils under chestnut stands in relation to previous land use

	Former agricultural land	Forest land	All
Soil depth (cm)	71.6	67.1	69.8
Stoniness (%)	27.4 a	35.5 b	30.3
pH KCl	4.3 (0.54)	4.2 (0.29)	4.2 (0.44)
Organic matter (%)	6.2 (2.1) a	8.4 (2.0) b	7.0 (3.3)
Total N (%)	0.26 (0.12) a	0.46 (0.11) b	0.36 (0.15)
C/N	9.6 (1.4) a	11.6 (2.3) b	10.4 (2.3)
Total S (%)	0.02 (0.02) a	0.04 (0.02) b	0.03 (0.02)
Extr. S (mg kg^{-1})	17.4 (9.5) a	9.3 (3.8) b	18.8 (9.0)
Extr. P (mg kg^{-1})	52.8 (65.5) b	3.8 (5.1) a	28.7 (52.6)
Extr. K (mg kg^{-1})	135.2 (100.7) b	76.8 (60.7) a	106.4 (87.9)
Extr. Ca (mg kg^{-1})	451.1 (517.1) b	221.3 (184.5) a	368.8 (415.5)
Extr. Mg (mg kg^{-1})	72.1 (67.1) b	26.5 (19.1) a	49.4 (54.4)
Extr. Mn (mg kg^{-1})	40.9 (24.1)	34.2 (25.1)	37.6 (24.6)
Extr. Fe (mg kg^{-1})	201.4 (74.3) b	138.8 (76.2) a	170.6 (81.1)
Extr. B (mg kg^{-1})	0.24 (0.10)	0.27 (0.12)	0.25 (0.11)
Extr. Zn (mg kg^{-1})	2.26 (1.86) b	1.20 (0.75) a	1.73 (1.51)
Extr. Cu (mg kg^{-1})	1.79 (1.18) b	0.63 (0.47) a	1.33 (1.11)
Extr. Ni (mg kg^{-1})	0.54 (0.45)	0.32 (0.18)	0.45 (0.38)

The values shown are means, with standard deviations in parenthesis. Different letters indicate significant differences

Table 5 Leaf weight and average foliar nutrient concentrations in the young chestnut stands studied

	Former agricultural land	Former forest land	All
Leaf weight (g leaf^{-1})	0.71 (0.16)	0.61 (0.20)	0.67
N (mg g^{-1})	27.8 (4.46) b	24.9 (4.50) a	26.76
S (mg g^{-1})	1.68 (0.40)	1.74 (0.35)	1.71
P (mg g^{-1})	1.86 (0.46) b	1.44 (0.47) a	1.65
P/N	0.06 (0.01) b	0.04 (0.02) a	0.05
Ca (mg g^{-1})	5.83 (2.52)	5.19 (2.27)	5.52
Ca/N	0.23 (0.10)	0.24 (0.10)	0.24
Mg (mg g^{-1})	2.69 (0.78)	2.35 (0.82)	2.53
Mg/N	0.17 (0.03)	0.21 (0.04)	0.18
K (mg g^{-1})	7.66 (1.74)	6.94 (1.21)	7.30
K/N	0.30 (0.08)	0.32 (0.09)	0.31
Al (mg g^{-1})	1.42 (0.43)	1.47 (0.63)	1.44
Al/Ca	0.26 (0.02)	0.32 (0.02)	0.28
Mn (mg g^{-1})	91.1 (63)	96.5 (110)	93.7
Fe (mg g^{-1})	112.1 (34)	114.7 (46)	113.4
Zn (mg g^{-1})	29.0 (8.5) b	21.8 (3.7) a	26.7
Cu (mg g^{-1})	4.6 (2.2) b	3.9 (2.3) a	4.3
Ni (mg g^{-1})	3.6 (2.9) b	2.9 (3.1) a	3.4

The values shown are means, with standard deviations in parenthesis. Different letters indicate significant differences

Table 6 Results of discriminant analysis for different levels of height (H10) and diameter growth (AID)

Predictor variables	Discriminant score[a]	Land use	Function	Positive variables	Negative variables	Sig.
Climatic	H10	Forest	1	ANDWP, ATP	ELEVATION, PD, ANDWF	0.003
	AID	Agric	1	AMT, SP	WP, AUMT, ANDWF	0.048
		Forest	1	ANDWOF, ATP, PD	ELEVATION, WP, PD	0.037
Foliar nutrient	H10	Forest	1	K, P	N, Al	0.033

[a] H10 is top height at a reference age of 10 years and AID is annual increment in dbh

and, to a lesser extent, foliar Mg concentrations were related to the corresponding soil extractable levels, with the relationships being more evident in the stands established on forest land (Pearson's correlation coefficient of 0.42 ($P = 0.014$) and 0.35 ($P = 0.048$), respectively). A slight relationship between pH and foliar levels of Ca (0.35, $P = 0.048$) was also found for forest land.

Site factors affecting tree growth

In the case of forest land, the discriminant analysis revealed discriminant functions that enabled classification of growth in relation to foliar concentrations or climatic variables. This pattern was, however, not observed in the stands established on former agricultural land (Table 6).

The effects of foliar nutrient concentrations on height and diameter growth in forest land are shown in Fig. 2 and Table 6. The lowest growth rates of trees were related to high foliar N and Al levels, whereas the stands with higher rates were characterized by high concentrations of K, P and Ca. Unlike plantations in forest areas, the effect of nutrition on growth in former agricultural land appeared to be less important, which may reflect a better overall nutritional status. In fact, only 20% of the 67 plots were deficient in P or Ca at such sites.

The analysis of the standardized discriminant coefficients in forest land revealed that elevation,

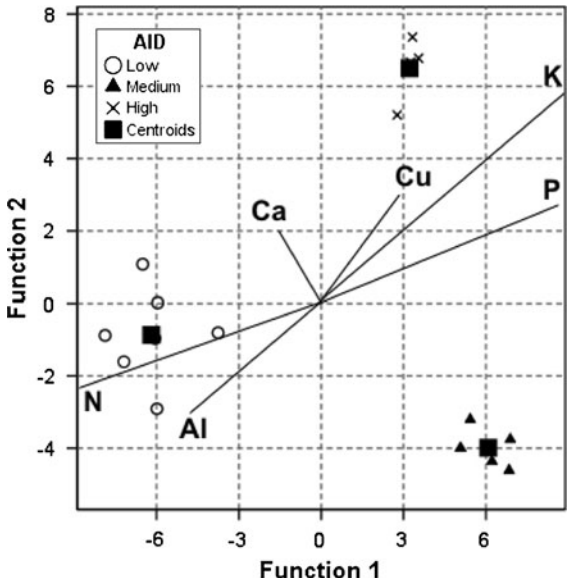


Fig. 2 Plots of discriminant functions in which foliar concentrations are used to classify AID in forest land

precipitation deficit and frost risk had a negative effect on height growth (Table 6). In contrast, total precipitation and number of days with precipitation were associated with better height growth. Poor conditions of diameter growth were associated with elevation, summer dryness and winter precipitation, probably due to soil leaching, as has been reported for other species (Moreno et al. 1996). On the contrary, improved diameter growth was associated with summer precipitation, annual precipitation and number of days without frost. In the case of agricultural land, the climatic conditions affected the diameter growth (Table 6), with winter precipitation, mean temperature in August and risk of frost all having negative effects, and the average mean temperature and summer precipitation positive effects. Continental sites with dry summers, rainy winters, long periods of frost and low average temperature, appeared to be unsuitable for timber production.

As stated in "Growth rates of chestnuts plantations" section, there were no significant differences between sweet and hybrid chestnut as regards growth rates. Nevertheless, it has to be taken into consideration that growth conditions and related factors were different in both cases, and consequently each type of chestnut may prosper under different growth conditions.

Discussion

Chestnut as a potential timber-producing species in plantations

The range obtained for top height at age 10 years was wide enough to include plantations with severe problems of early establishment (H10 $<$ 6 m), and plantations with very good growth conditions from early ages (H10 $>$ 8 m). This can be compared with the average value of 5 m top height at 10 years recorded in the classic Spanish yield tables (Ximénez de Embún 1963), and with the mean value of 8.3 m for sites in the Forest of Dean, in southern England, although the growth pattern appears to be more sustained with age in this case (Everard and Christie 1995). The slight decrease in height growth indicated by the Galician curves requires further confirmation, but is consistent with the changes in top height in mixed Douglas fir/chestnut plantations in northern Portugal (Sales and Monteiro 1998).

The mean annual increment in diameter is also a good indicator of the productive potential of these chestnut plantations. The mean value for plots aged 7–16 years was 0.90 cm year^{-1}, which is consistent with the potential for growing trees of 40 cm diameter on a rotation of 40–50 years (Bourgeois 1992). Much better results could be obtained if clones selected specifically for height growth and aptitude for timber production were used in monoclonal plantations.

Site requirements for plantations destined for timber production

Several ecological site classifications in use in Europe assume that the three principal factors needed to describe a site are climate, soil moisture and soil nutrient regime (Pyatt et al. 2001). Analysis of site requirements for different species according to these factors can help silviculturists to select the correct species and even provenances for a site (Kerr and Cahalan 2004).

Chestnut growth decreases with elevation as a direct effect of the lack of warmth: this limitation is more obvious in forest areas, since agricultural land is generally sited at lower elevations. The annual number of days without frost, evapotranspiration, summer precipitation and precipitation deficit appear

as the key climatic factors in considering the site requirements of this species, with effects associated with annual mean temperature (positive) and August mean temperature (negative) in the case of agricultural land. In other Mediterranean regions of Spain, the length of drought period has also been identified as a main climatic limitation for growth of chestnut (Rubio et al. 2002; Gandullo Gutiérrez et al. 2004), but the thresholds defined for the whole of Spain or Catalonia do not appear to be directly applicable to the moisture-demanding timber production plantations. For the latter, we propose the following threshold values for 'optimal' growth conditions: a period free of risk of frost of at least 3-4 months, evapotranspiration (ETP) < 650 mm, summer precipitation (SP) > 130 mm, precipitation deficit (PD) < 200 mm and elevation lower than 800 m.

In the west of Spain (Extremadura), the climate can be characterized by a significant autumn and spring rainfall regime (334 and 329 mm, respectively), warm mean annual temperatures (13°C) and high summer evapotranspiration rates. In this Mediterranean Basin, Rubio et al. (1997) showed that the presence of chestnut coppice stands for timber production depended on the amount and distribution of precipitation, mainly during the summer. For this reason, chestnut coppice has been restricted to the most climatically favoured zones. The relevance of summer drought in this region has also been indicated.

The chemical soil properties in most of the plantations corresponded to the central 80% portion of the chestnut habitat in Spain, as defined by Gandullo Gutiérrez et al. (2004), but values higher than the upper thresholds were found for the organic matter and soil total nitrogen, which is consistent with results obtained in regional studies (Blanco et al. 2000). Although some plantations were growing on soils that were more acidic and richer in organic matter than the range observed for the Spanish chestnut habitat, tree growth was not depleted. The results suggest the lack of strong limitations of the agricultural soils in which plantations were established, in contrast with other species in the region (Merino et al. 2003). Chestnut appears to be more accommodating as regards soil nutrient regime than previously considered, although most of the plots had not yet reached canopy closure, and thus nutrient requirements of the stand were high compared with biomass production (Ranger and Colin-Belgrand 1996).

Regional studies have been carried out in Spain under fairly homogeneous climatic conditions to quantify some of the main edaphic and physiographic characteristics of the chestnut stands. These types of studies are of interest as regards the information obtained about the autoecology of the chestnut tree in Spain. Comparison of the most important edaphic habitat characteristics of this species in Galicia (Blanco et al. 2000; Rubio et al. 2001), Catalonia (Rubio et al. 2002) and Extremadura (Rubio et al. 1997) revealed that: (i) Galician chestnut occurs preferentially at altitudes between 400 and 750 m, while in northeastern and west Iberian Peninsula it may reach higher altitudes, in all cases on very stony land and very varied slopes (both of these factors cause low water holding capacity in soils); (ii) in all areas the soils are developed on acid bedrock, with forest mull humus forms, and moderate contents of organic matter in Galicia and Extremadura, and poor contents in Catalonia; and (iii) in Galicia and Catalonia the predominant texture in soils is loamy-sand, and in Extremadura, loam texture predominates.

Although European chestnut is known to grow on naturally acid soils, nutrient deficiencies may occur as a consequence of poor nutrient reserves or nutrient imbalances in the soil, leading to detrimental effects on tree growth and nut production (Portela et al. 2003). The concentrations of P and K were positively related to growth and were significantly lower in former forest areas, as expected from the results of previous studies (Laroche et al. 1997). The plantations in this study displayed lower foliar concentrations of P and K, similar values for Ca and Mg and much higher foliar N than other chestnut stands with different structures and management (coppice stands, orchards), in other regions of Spain and Portugal (Santa Regina 2000; Pires et al. 1995). Foliar N and growth were negatively related in forest land, as also observed for other species planted in the area (Sánchez Rodríguez et al. 2002). The micronutrients analyzed were within the ranges previously recorded in healthy young chestnut stands.

Conclusions

Differences in growth of the plantations under study are largely attributed to the former use of the land, which determined the soil characteristics, although

these may also be related to local or regional climatic conditions. The results indicate the need for appropriate choice of species and the possibility of growing both hybrid and sweet chestnut with a high rate of early growth. Deep soil preparation is essential for forest land, in which fertilization is also required, with repeated applications of high doses of fertilizer apparently necessary. Higher summer precipitation, lower elevation and absence of long periods of frost risk provide the best growth conditions in Galicia.

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